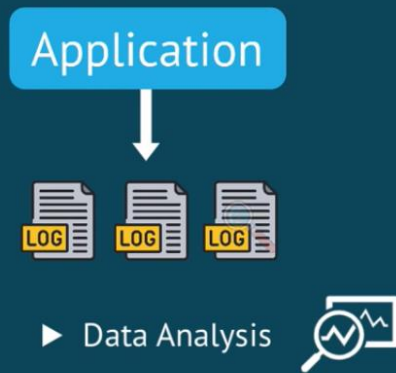


Fluent Bit (Log and Metrics Processor)

Why we need log data?



Challenges of Logging



Mobile & Web App Logs



Access Logs



System Logs



Databases



Challenges of Logging



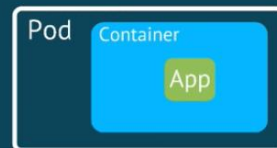
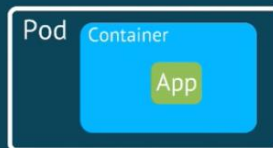
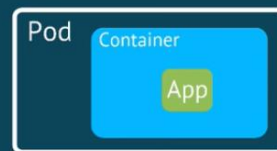
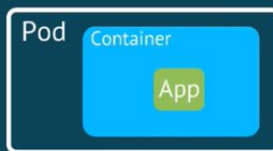
Gather metrics:

▶ CPU

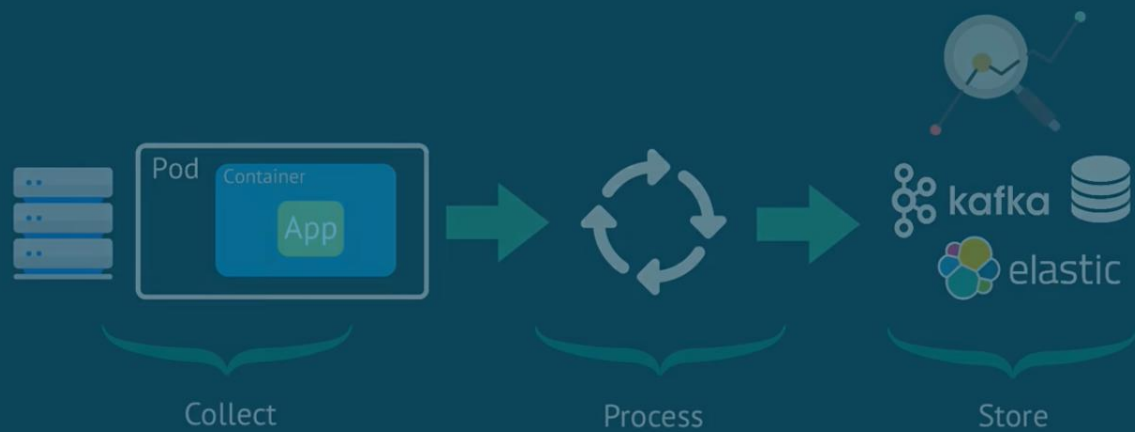
▶ Memory

▶ Storage

Challenges of Logging in Kubernetes



Challenges of Logging in Kubernetes



Challenges of Logging in Kubernetes

Resource Requirement

Application Performance vs.
Log Mechanism



How does Fluent Bit work?



Input Plugins

- ▶ gather information from different sources

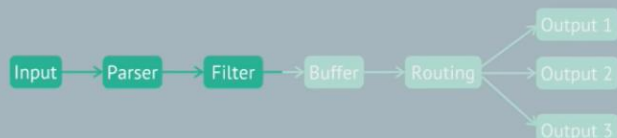


How does Fluent Bit work?



Parsers

- ▶ convert unstructured to **structured** messages



How does Fluent Bit work?



Parsers

- convert unstructured to **structured** messages

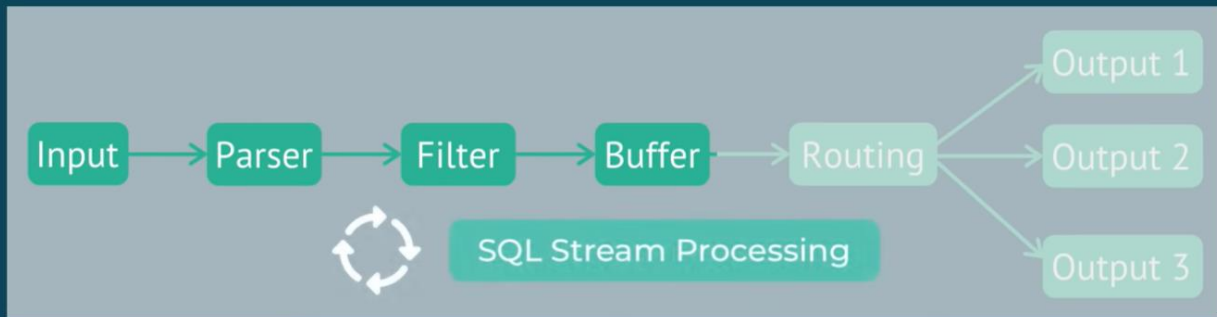


Filters

- **modify** your records
- **enrich** your records

```
{
  "log": "Hi Fluent Bit",
  "stream": "stdout",
  "time": "2021-04-22T08:05:12.235654328Z",
  "kubernetes": {
    "host": "minikube",
    "pod_name": "MyApp",
    "pod_id": "1231244",
    "container_name": "MyApp"
  }
}
```

How does Fluent Bit work?



How does Fluent Bit work?



SQL Stream Processing

- ▶ SQL Queries on logs or metrics

Aggregations

Calculations

Time Series Predictions

- ▶ Calculate AVG, MAX, MIN, COUNT, ...

How does Fluent Bit work?



SQL Stream Processing

- ▶ SQL Queries on logs or metrics

Aggregations

Calculations

Time Series Predictions

- ▶ No database and no indices required 

How does Fluent Bit work?



Output Plugins

- ▶ define destinations for the data



splunk>



How does Fluent Bit work?

TCP Format

Fluent Bit Format

ElasticSearch Format

Input
Plugins



Output
Plugins

- How to transform data to **Fluent Bit** format

- How to transform data to **output target** format

Flexible Routing

Two main concepts: Tags and Matching

```
[INPUT]
  Name apache
  Tag  apache.data
  ...

[FILTER]
  Name parser
  Match apache.*
```

- Tag **identifies** the data source



TAGS



MATCHING

- Match against the **tags** of incoming records



Master

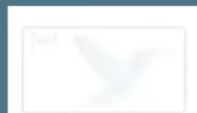
Fluent Bit in Kubernetes



Node 1



Node 2



Node 3

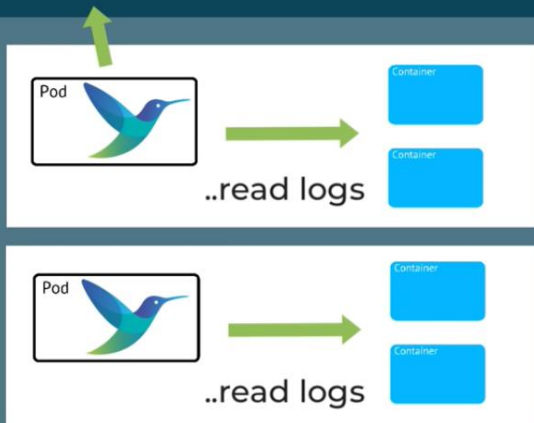


- ▶ Deployed as DaemonSet
- ▶ Runs on every Node



Master

Fluent Bit in Kubernetes



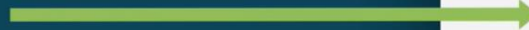
- ▶ Reads logs from containers
- ▶ Gather metadata from K8s API Server

Advantages of Fluent Bit



Kubernetes Annotations

Suggest a Parser



```
apiVersion: v1
kind: Pod
metadata:
  name: apache-logs
  labels:
    app: apache-logs
  annotations:
    fluentbit.io/parser: apache
spec:
  containers:
  - name: apache
    image: edsiper/apache_logs
```

Advantages of Fluent Bit



Built-In Security

- ▶ Send logs safely to storage backends
- ▶ TLS connections



Easy Scalable

- ▶ Simple architecture